

Short-time asymptotics for solutions of the evolutionary game theoretic p -laplacian and Pucci operators

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Let $u = u(x, t)$ the solution to the following initial-boundary value problem:

$$u_t - F(\nabla u, \nabla^2 u) = 0 \quad \text{in } \Omega \times (0, \infty), \quad (1)$$

$$u = 0 \quad \text{on } \Omega \times \{0\}, \quad (2)$$

$$u = 1 \quad \text{on } \Gamma \times (0, \infty). \quad (3)$$

where Ω is a domain in $\mathbb{R}^N$, with $N \geq 2$, with (not necessarily bounded) boundary $\Gamma \neq \emptyset$.

Here, F is a generic operator that will be specified in what follows. Since we shall consider nonlinear operators not in divergence form we shall intend (1) satisfied in according to the **theory of viscosity solutions**.

When Ω is unbounded we mean that we shall consider the bounded solution u .

We shall focus on the **short-time behavior** of u .

The linear case – Varadhan formula

Theorem ([Varadhan, 1967])

Let u be the solution to $u_t - \Delta u = 0$ in $\Omega \times (0, \infty)$, satisfying (2) and (3). Then, we have that

$$-4 \lim_{t \rightarrow 0^+} t \log u(x, t) = d_\Gamma(x)^2 \quad \text{for any } x \in \overline{\Omega}. \quad (4)$$

Here, for $x \in \Omega$, $d_\Gamma(x)$ is the **distance of x from the boundary Γ** :

$$d_\Gamma(x) = \inf\{|y - x| : y \in \Gamma\}.$$

Formula (4) holds when one merely asks that $\Gamma = \partial(\mathbb{R}^N \setminus \Omega)$ and it is satisfied uniformly on every compact subset of $\overline{\Omega}$.

The linear case – Formula for the heat content

Magnanini and Sakaguchi gave the following result (first in [*Proc Roy. Soc. Edimburgh*, 2007] and then in [*J. Differential Equations*, 2012] for the case of domains with unbounded boundary).

Theorem (Magnanini, Sakaguchi)

Let $x \in \Omega$ and assume that the open ball $B_R(x)$ centered at x and with radius R is contained in Ω and such that $\overline{B_R(x)} \cap \Gamma = \{y_x\}$ for some $y_x \in \Gamma$. Then we have:

$$\lim_{t \rightarrow 0^+} t^{-\frac{N+1}{4}} \int_{B_R(x)} u(z, t) dz = C_N \left(\prod_{j=1}^{N-1} \left[\frac{1}{R} - \kappa_j(y_x) \right] \right)^{-\frac{1}{2}},$$

where $\kappa_1, \dots, \kappa_{N-1}$ are the principal curvatures of Γ at y_x and C_N is a positive constant.

The linear case – Formula for the heat content

The formula holds true even if $k_j(y_x) = \frac{1}{R}$ for some $j = 1, \dots, N-1$ and it is satisfied if the domain is smooth only in a neighborhood of y_x .

Formula for the mean value

From (4) descends the following scale invariant formula for the mean value:

$$\lim_{t \rightarrow 0^+} \left(\frac{R^2}{t} \right)^{\frac{N+1}{4}} \oint_{B_R(x)} u(z, t) dz = C_N \left(\prod_{j=1}^{N-1} [1 - R\kappa_j(y_x)] \right)^{-\frac{1}{2}}. \quad (5)$$

Our main contributions

Taking advantage on new barriers we have been able to **generalize Formulas (4) and (5)** to the nonlinear diffusion cases of either Δ_p^G or $\mathcal{M}^\pm$. Moreover, we **have extended** these results to new **sharp estimates on the rate of convergence of (4) and asymptotic formulas for the q -means** (that are new also in the linear case).

These results are contained in the following works:

- [B.-Magnanini, *J. Math. Pures Appl.*, 2019, 1709.10005];
- [B.-Magnanini, *Applicable Analysis*, 2019, 1801.04230];
- [B. (Ph.D. thesis) 2019, 1902.10346];
- [B.-Magnanini (in preparation)].

Game-Theoretic p -Laplacian

The **game-theoretic p -laplacian** is defined by, for $1 < p < \infty$,

$$\Delta_p^G u = \frac{1}{p} |\nabla u|^{2-p} \Delta_p u$$

where Δ_p is the usual p -laplacian

$$\Delta_p u = \operatorname{div}(|\nabla u|^{p-2} \nabla u)$$

and in the extremal case $p = \infty$ by

$$\Delta_\infty^G u = \left(\nabla^2 u \frac{\nabla u}{|\nabla u|} \cdot \frac{\nabla u}{|\nabla u|} \right).$$

Note that, for any $1 < p \leq \infty$ we formally have that

$$\Delta_p^G u = \frac{\Delta u}{p} + \left(1 - \frac{2}{p} \right) \Delta_\infty^G u.$$

Essential properties of GT p -Laplacian. When $p = 2$ we obtain half of the usual Laplacian, $\Delta_2^G u = \frac{1}{2} \Delta u$.

It is not linear but it is **1-homogeneous**, that is

$$\Delta_p^G(\lambda u) = \lambda \Delta_p^G u \quad \text{per ogni } \lambda \in \mathbb{R},$$

differently from Δ_p , which is $(p - 1)$ -homogeneous.

As quasi-linear operator Δ_p^G can be seen as

$$\Delta_p^G u = \operatorname{tr} \left(A(\nabla u) \nabla^2 u \right),$$

where the coefficients A are discontinuous of if $\nabla u = 0$.

Δ_p^G is **uniformly elliptic** if $p \in (1, \infty)$ and **(degenerate) elliptic** in the case $p = \infty$. Formally, we have that for any $\xi \in \mathbb{R}^N \setminus \{0\}$,

$$\min \left\{ \frac{1}{p}, \frac{1}{p'} \right\} |\eta|^2 \leq (A(\xi) \eta \cdot \eta) \leq \max \left\{ \frac{1}{p}, \frac{1}{p'} \right\} |\eta|^2, \quad \text{for any } \eta \in \mathbb{R}^N$$

where $p' = p/(p - 1)$ if $1 < p < \infty$ and $p' = 1$ if $p = \infty$.

Pucci operators

Given $0 < \lambda \leq \Lambda < \infty$, setting $\mathcal{A}_{\lambda,\Lambda}$ the set of symmetric matrices with eigenvalues included in $[\lambda, \Lambda]$.

The **Pucci extremal operators** $\mathcal{M}^\pm$ are defined by:

$$\mathcal{M}^+(X) = \sup \{ \operatorname{tr}(AX) : A \in \mathcal{A}_{\lambda,\Lambda} \}$$

and

$$\mathcal{M}^-(X) = \inf \{ \operatorname{tr}(AX) : A \in \mathcal{A}_{\lambda,\Lambda} \}.$$

Equivalently, it is easy to see that $\mathcal{M}^\pm$ can be seen as

$$\mathcal{M}^+(X) = \lambda \sum_{e_i < 0} e_i + \Lambda \sum_{e_i > 0} e_i$$

and

$$\mathcal{M}^-(X) = \lambda \sum_{e_i > 0} e_i + \Lambda \sum_{e_i < 0} e_i$$

where $e_i = e_i(X)$ for $i = 1, \dots, N$ are the eigenvalues of X .

Essential properties of Pucci operators. The Pucci operators are fully nonlinear operators. They **generalize** the classical Laplacian, which is obtained when $\lambda = \Lambda$. They are **positively one-homogeneous**:

$$\mathcal{M}^{\pm}(\nabla^2(\delta u)) = \delta \mathcal{M}^{\pm}(\nabla^2 u) \quad \text{for all } \delta \geq 0.$$

Although equations for Pucci operators are well studied in terms of $W^{2,N}$ solutions, they well fit also the theory of viscosity solutions. Indeed, they are **(degenerate) elliptic in the sense of viscosity solutions**, i.e.

$$\mathcal{M}^{\pm}(X) \geq \mathcal{M}^{\pm}(Y) \quad \text{if } X \geq Y.$$

To unify the presentation of the results, we introduce the symbol α that has the role of parameter depending on which operator we shall take into account.

Let α be the equals to p' for the GT p -laplacian, to $\frac{1}{\lambda}$ in the case of $\mathcal{M}^+$ and to $\frac{1}{\lambda}$ for $\mathcal{M}^-$.

Recall that given $p \in (1, \infty]$,

$$p' = \begin{cases} \frac{p}{p-1} & \text{se } p \in (1, \infty) \\ 1 & \text{se } p = \infty. \end{cases}$$

Hence, we state the first result that we have obtained, which generalizes (4).

Theorem ([B.-Magnanini, JMPA],[B.-Magnanini, in preparation])

Let Ω be a domain such that $\Gamma = \partial(\mathbb{R}^N \setminus \overline{\Omega})$. Let u be the solution to (1)-(3). Then, it holds that

$$\lim_{t \rightarrow 0^+} 4t \log u(x, t) = -\alpha d_\Gamma(x)^2, \quad x \in \overline{\Omega}. \quad (6)$$

In what formula (6) is based on? New barriers argument that employs radial solutions of the equation (1). Accurate estimates on the asymptotic behavior in these radial cases and a covering argument relied on comparison and maximum principles produce the result.

This is due to the fact that, for radially symmetric functions $u(x, t) = u(|x|, t)$, either Δ_p^G acts as a linear operator and $\mathcal{M}^\pm$ has a piecewise linear structure.

Also, a key step is the revealed linear-like link between the solution u and the solution u^ε of the auxiliary problem:

$$u - \varepsilon^2 F(\nabla u, \nabla^2 u) = 0 \quad \text{in } \Omega \quad (7)$$

$$u = 1 \quad \text{on } \Gamma. \quad (8)$$

In particular, we make use of the asymptotic profile of u^ε for small $\varepsilon \rightarrow 0^+$.

Theorem ([B.-Magnanini, AA], [B.-Magnanini, in preparation])

Assume that $\Gamma = \partial(\mathbb{R}^N \setminus \overline{\Omega})$. Let u^ε be the solution to (9)-(10). Then, it holds that

$$\lim_{\varepsilon \rightarrow 0^+} \varepsilon \log u^\varepsilon(x) = -\sqrt{\alpha} d_\Gamma(x), \quad x \in \overline{\Omega}.$$

As it has been mentioned, barrier functions based on **radial solutions** are employed. In particular from the global solution of (1) in $\mathbb{R}^N \times (0, \infty)$ we construct barriers from below. The barrier from above on the other hand are built from the solution u^ε of the auxiliary problem:

$$u - \varepsilon^2 F(\nabla u, \nabla^2 u) = 0 \quad \text{in } \Omega \quad (9)$$

$$u = 1 \quad \text{on } \Gamma. \quad (10)$$

In particular, we make use of the asymptotic profile of u^ε for small $\varepsilon \rightarrow 0^+$.

Theorem ([B.-Magnanini, AA], [B.-Magnanini, in preparation])

Assume that $\Gamma = \partial(\mathbb{R}^N \setminus \overline{\Omega})$. Let u^ε be the solution to (9)-(10). Then, it holds that

$$\lim_{\varepsilon \rightarrow 0^+} \varepsilon \log u^\varepsilon(x) = -\sqrt{\alpha} d_\Gamma(x), \quad x \in \overline{\Omega}.$$

Asymptotics for q -means

Given $x \in \Omega$, $t > 0$ and $q \in [1, \infty]$, the q -**mean** $\mu_q(x, t)$ on $B_R(x) \subset \Omega$ of u the solution to (1)-(3) is the unique real number μ such that

$$\|u(\cdot, t) - \mu\|_{L^q(B_R(x))} \leq \|u(\cdot, t) - \lambda\|_{L^q(B_R(x))} \quad \text{for any } \lambda \in \mathbb{R}.$$

Remark

μ_q generalize the **mean value** of u on $B_R(x)$, which is obtained when one considers $q = 2$ in the previous definition:

$$\mu_2(x, t) = \int_{B_R(x)} u(z, t) \, dz.$$

Asymptotics for q -means

Theorem ([B.-Magnanini, JMPA], [B.-Magnanini, in prep.])

Let $x \in \Omega$ and $R > 0$ such that $\overline{B_R(x)} \cap \Gamma = \{y_x\}$ for some $y_x \in \Gamma$. Let u be the solution to (1)-(3). Given $q \in (1, \infty]$, let $\mu_q(x, t)$ the q -mean of u on $B_R(x)$.

Then, for any $q \in (1, \infty)$, we have that

$$\lim_{t \rightarrow 0^+} \left(\frac{R^2}{t} \right)^{\frac{N+1}{4(q-1)}} \mu_q(x, t) = C_{N,q} \left(\frac{\alpha^{\frac{N+1}{2}}}{\prod_{j=1}^{N-1} [1 - R \kappa_j(y_x)]} \right)^{\frac{1}{2(q-1)}}, \quad (11)$$

where $C_{N,q}$ is a positive constant depending only on the labeled parameters.

Notice the role of α and q .

(11) **generalizes** to every $p \in (1, \infty]$ and **extend** to any $q \in (1, \infty)$, formulas obtained by Magnanini e Sakaguchi in the case $p = q = 2$. In the extremal case $q = \infty$, we have obtained that

$$\mu_\infty(x, t) \rightarrow \frac{1}{2} \text{ per } t \rightarrow 0^+.$$

Key points on which the proof of (11) is based on

- The behavior for small values of $s > 0$ of

$$\mathcal{H}_{N-1}(\Gamma_s \cap B_R(x))$$

in terms of the principal curvatures. Here, $\Gamma_s = \{x \in \Omega : d_\Gamma(x) = s\}$, for $s > 0$ and $\mathcal{H}_{N-1}$ is the $(N - 1)$ -dimensional Hausdorff measure.

- Uniform sharp estimates (new also in the linear case) on the rate of convergence of the Varadhan-type formulas (6).

Uniform sharp estimates

If Ω is of class $C^{0,\omega}$, i.e. Γ is locally a graph of a continuous function whose modulus of continuity is controlled by ω .

Theorem (Uniform sharp estimates [B.-Magnanini, JMPA],[B.-Magnanini, in preparation])

Let $\Omega \in C^{0,\omega}$. Let u be the solution to (1)-(3). Then, we have that

$$4 t \log u(x, t) + \alpha d_{\Gamma}(x)^2 = O(t \log \psi_{\omega}(t)),$$

for $t \rightarrow 0^+$, uniformly on compact subsets of $\overline{\Omega}$.

The function $\psi_{\omega}(t)$ depends on ω , is positive and vanishes for $t \rightarrow 0^+$.

Remark

If $\Omega \in C^{0,\alpha}$, with $0 < \alpha < 1$, then $O(t \log \psi_{\omega}(t)) = O(t \log t)$, for $t \rightarrow 0^+$.

The elliptic counterparts

Theorem (Uniform estimates for elliptic GT p -Laplacian [B.-Magnanini, AA])

Let $\Omega \subset \mathbb{R}^N$ and u^ε be the solution to (9)-(10). If Ω is such that $\Gamma = \partial(\mathbb{R}^N \setminus \overline{\Omega})$, then we have that:

$$\varepsilon \log u^\varepsilon(x) + \sqrt{p'} d_\Gamma(x) = \begin{cases} O(\varepsilon) & \text{if } p = \infty, \\ O(\varepsilon \log \varepsilon) & \text{if } p > N, \end{cases}$$

for $\varepsilon \rightarrow 0^+$, on every compact subset of $\overline{\Omega}$.

If $\Omega \in C^{0,\omega}$, it holds that

$$\varepsilon \log u^\varepsilon(x) + \sqrt{p'} d_\Gamma(x) = \begin{cases} O(\varepsilon \log |\log \psi_\omega(\varepsilon)|) & \text{if } p = N, \\ O(\varepsilon \log \psi_\omega(\varepsilon)) & \text{if } 1 < p < N, \end{cases}$$

for $\varepsilon \rightarrow 0^+$, uniformly on every compact subset of $\overline{\Omega}$.

Remark

- The **critical threshold for p** in the last formula seems to be related to the integrability on the variable t of the global solution of (1).
- Also, we observe that, (8), (10) and (10) can be easily extended when is changed the **boundary data** provided that it is still finite and far away from zero.

Theorem ([B.-Magnanini, in preparation])

Assume $\Omega \in C^{0,\omega}$. Let u^ε be the solution to $u^\varepsilon - \varepsilon^2 \mathcal{M}^-(\nabla^2 u) = 0$ in Ω such that $u^\varepsilon = 1$ on Γ . Then, it holds that for $\varepsilon \rightarrow 0$ uniformly on compact subsets of $\overline{\Omega}$:

$$\varepsilon \log u^\varepsilon + \frac{d_\Gamma}{\sqrt{\lambda}} = \begin{cases} O(\varepsilon \log |\log \psi_\omega(\varepsilon)|) & \text{if } N = 2 \text{ and } \lambda = \Lambda, \\ O(\varepsilon \log \psi_\omega(\varepsilon)) & \text{if } N \neq 2 \text{ or } \lambda \neq \Lambda. \end{cases}$$

Let u^ε be the solution to $u^\varepsilon - \varepsilon^2 \mathcal{M}^+(\nabla^2 u) = 0$ in Ω such that $u^\varepsilon = 1$ on Γ . Then, it holds that for $\varepsilon \rightarrow 0$ uniformly on compact subsets of $\overline{\Omega}$:

$$\varepsilon \log u^\varepsilon + \frac{d_\Gamma}{\sqrt{\Lambda}} = \begin{cases} O(\varepsilon \log \varepsilon) & \text{if } \Lambda > \lambda(N-1), \\ O(\varepsilon \log |\log \psi_\omega(\varepsilon)|) & \text{if } \Lambda = \lambda(N-1), \\ O(\varepsilon \log \psi_\omega(\varepsilon)) & \text{if } \Lambda < \lambda(N-1). \end{cases}$$

If it is considered the dimensional-like quantity $\tilde{N}_+ = \frac{\lambda}{\Lambda}(N-1) + 1$, then the previous estimate can be restated as

$$\varepsilon \log u^\varepsilon + \frac{d_{\Gamma}}{\sqrt{\Lambda}} = \begin{cases} O(\varepsilon \log \varepsilon) & \text{if } \tilde{N}_+ < 2, \\ O(\varepsilon \log |\log \psi_\omega(\varepsilon)|) & \text{if } \tilde{N}_+ = 2, \\ O(\varepsilon \log \psi_\omega(\varepsilon)) & \text{if } \tilde{N}_+ > 2. \end{cases}$$

The quantity $\tilde{N}_+$ is related to the critical exponent q , in the sense that the equation ($N \geq 3$)

$$\mathcal{M}^+(\nabla^2 u) + u^q \leq 0 \text{ in } \mathbb{R}^N \text{ and } u \geq 0 \text{ in } \mathbb{R}^N$$

has a nontrivial solution if and only if $1 < q < \frac{\tilde{N}_+}{\tilde{N}_+ - 2}$.

Thanks for your kind attention!